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Studierichting: Informatica
Oefeningenreeks 4A nr. 43.5

$$I = \int \frac{3x^4 - 6x^2 - 4x + 4}{x^5 - 3x^3 + 2x^2} dx = \int \frac{3x^4 - 6x^2 - 4x + 4}{x^2(x-1)^2(x+2)} dx$$

$$\frac{3x^4 - 6x^2 - 4x + 4}{x^2(x-1)^2(x+2)} = \frac{A}{x^2} + \frac{B}{x} + \frac{C}{(x-1)^2} + \frac{D}{x-1} + \frac{E}{x+2}$$

$$A = \left. \frac{3x^4 - 6x^2 - 4x + 4}{(x-1)^2(x+2)} \right|_{x=0} = 2$$

$$\left(\frac{3x^4 - 6x^2 - 4x + 4}{x^2(x-1)^2(x+2)} - \frac{2}{x^2} \right) x = \frac{3x^4 - 6x^2 - 4x + 4 - 2(x-1)^2(x+2)}{x(x-1)^2(x+2)}$$

$$= \frac{3x^4 - 6x^2 - 4x + 4 - 2x^3 + 8x - 2x - 4}{x(x-1)^2(x+2)} = \frac{x(3x^3 - 2x^2 + 2)}{x(x-1)^2(x+2)} = \frac{3x^3 - 2x^2 + 2}{(x-1)^2(x+2)}$$

$$\Rightarrow B = \left. \frac{3x^3 - 2x^2 + 2}{(x-1)^2(x+2)} \right|_{x=0} = 1$$

$$C = \left. \frac{3x^4 - 6x^2 - 4x + 4}{x^2(x+2)} \right|_{x=1} = -1$$

$$\left(\frac{3x^4 - 6x^2 - 4x + 4}{x^2(x-1)^2(x+2)} + \frac{1}{(x-1)^2} \right) (x-1) = \frac{3x^4 - 6x^2 - 4x + 4 + x^2(x+2)}{x^2(x-1)(x+2)}$$

$$= \frac{3x^4 - 6x^2 - 4x + 4 + x^3 + 2x^2}{x^2(x-1)(x+2)} = \frac{3x^4 + x^3 - 4x^2 - 4x + 4}{x^2(x-1)(x+2)}$$

$$= \frac{(x-1)(3x^3 + 4x^2 - 4)}{x^2(x-1)(x+2)} = \frac{3x^3 + 4x^2 - 4}{x^2(x+2)}$$

$$\Rightarrow D = \left. \frac{3x^3 + 4x^2 - 4}{x^2(x+2)} \right|_{x=1} = 1$$

$$E = \left. \frac{3x^4 - 6x^2 - 4x + 4}{x^2(x-1)^2} \right|_{x=-2} = 1$$

$$\Rightarrow I = \int \frac{2dx}{x^2} + \int \frac{dx}{x} + \int \frac{-dx}{(x-1)^2} + \int \frac{dx}{x-1} + \int \frac{dx}{x+2}$$

$$= \frac{-2}{x} + \ln|x| + \frac{1}{x-1} + \ln|x-1| + \ln|x+2| + C$$

References